

Assignment: Deconvolution and Image Transformations

Signal and Image Processing

February 25, 2026

You can work on this assignment and submit your solution (report and code) as a GROUP. This assignment counts towards your grade and have to be submitted in order to pass the course. You must follow the report guidelines found in `guidelines.pdf`. The page limit for this assignment is **15 pages** including everything, i.e. illustrations and code snippets.

1. Transformations on images - Translation

Let $I(x, y)$ be a 2D image. Consider the image $\tilde{I}$ obtained by translation of I by one pixel to the right.

- 1.1. (1 point) Express what $\tilde{I}(x, y)$ is with respect to the pixels in image I .

Deliverables: A mathematical expression.

- 1.2. (1 point) Write the matrix corresponding to this transformation expressed in homogeneous coordinates and write how this transformation can be used to produce $\tilde{I}$.

Deliverables: Two mathematical expressions.

- 1.3. (1 point) It is possible to formulate this translation using a linear filter. Write the linear filter kernel corresponding to this transformation.

Deliverables: Include the filter kernel in the report.

- 1.4. (1 point) Create a zero-valued image of odd size in x and y containing a centred white square.

Deliverables: A figure illustrating the image.

- 1.5. (1 point) Write a Python function to translate this image by any specified integer number of pixels in both directions using filter masks like in question 1.3. You can use any existing Python package to do the filtering.

Deliverables: Include your code for the function in the report and a figure illustrating the results of applying the function to the image from the previous question. Discuss the different boundary conditions that you can put and what they do.

- 1.6. (1 point) Make a new function to handle non-integer translations, using the homogeneous transform formulation from question 1.2. and nearest neighbor interpolation. You are not allowed to use `skimage.transform.warp` or similar – instead you should implement this simple image transformation yourself.

Deliverables: Include your code for the function in the report and an illustration of your white square image, as well as the image you get by translating by $\mathbf{t} = (0.6, 1.2)^T$.

- 1.7. (1 point) Recall the equation for the Fourier transform of a translated image with respect to the Fourier transform of the original image. Write a Python function that performs translation based on the Fourier transform and compare outputs with the previous approach.

Deliverables: Include your code for the function in the report and a figure illustrating the results of applying the function to your white square image. Include a few sentences of comparison with the previous approaches.

- 1.8. (1 point) Can you do non-integer translations with the Fourier method? Comment on what happens for the previously defined white square image. Apply it also on another image.

Deliverables: Answer to the question and comments about the white square image and examples of applying it to another image of your choice.

2. Feature detection and image transforms

- 2.1. (2 point) The image `textlabel_gray_small.png` contains a label with some text on which we would like to transform such that the label is horizontal in order for better reading of the text (e.g. necessary if we want to pass it to an optical character recognition (OCR) algorithm). It is not enough to simply rotate the image 90 degrees clockwise. Implement in Python an automatic method for detecting the corners of the label and use these to estimate a transformation that estimate the proper rotation of the image. You can use your Harris corner detector from Week 2, but you need to adjust the parameters carefully and some heuristics to find the label corners from the points detected by the Harris corner detector. Hint: Maybe you only need two opposing sides of the label in order to estimate the required rotation. Once you have corner points, you can compute an angle needed for rotation around the center of the image and use `skimage.transform.rotate` to rotate the label image.

Deliverables: Briefly describe the essential steps in your solution including relevant code snippets. Remember to specify all relevant parameters in the report. Include illustrations of the detected label corner points overlayed on top of the image and the resulting rotated image and state your estimated rotation angle.

3. Features, filter banks and texture

- 3.1. (1 point) Designed filter bank - Implement a filterbank based on scale normalized derivatives of the Gaussian (N-Jet Filter Bank) up to and including third order at multiple scales. Load the `files->Material->Test Images->Week 1->sunandsea.jpg` image in grayscale and visualize the filters as well as the results of filtering the image with the filter bank. **Deliverables:** Briefly describe the essential steps in the solution and include a plot showing visualizations of the filters and the filter responses

- 3.2. (1 point) Learned filter bank - Use PCA to learn filters from patches of the `sunandsea.jpg` image. Try for instance patches of 8×8 in size or experiment with different sizes. You can implement your own patch sampling function or use

`sklearn.feature_extraction.image.extract_patches_2d`. Visualize the filters as well as the results of filtering the image with the filter bank. Compare and discuss your observations with the results of the N-Jet Filter Bank. **Deliverables:** Briefly describe the essential steps in the solution and include a

plot showing visualizations of the filters and the filter responses. Include a few sentences on the comparison/discussion.

- 3.3. (1 point) Segmentation with designed and learned features - Implement K-Means clustering (with $K = 3$) based on the above two implemented filter banks and use it to do unsupervised segmentation of the image. Remember, K-Means is sensitive to the number of dimensions, so you may need to limit the number of features you use. For the N-Jet Filter Bank, you can for instance use PCA to limit the number of dimensions and/or consider rotation invariant versions by computing the maximum over each directional derivative. What do you observe? Can you arrive at reasonable segmentation and which filter bank perform the best? **Deliverables:** Briefly describe the essential steps in the solution and include a plot showing visualizations of the segmentation. Include a few sentences on the observations and conclusions.